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Higher ratio of plasma omega-6/omega-3 fatty acids is associated with greater risk of all-cause, cancer, and cardiovascular mortality: a population-based cohort study in UK Biobank

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Abstract

Background

Circulating omega-3 and omega-6 polyunsaturated fatty acids (PUFAs) have been associated with various chronic diseases and mortality, but results are conflicting. Few studies examined the role of omega-6/omega-3 ratio in mortality.

Methods

We investigated plasma omega-3 and omega-6 PUFAs and their ratio in relation to all-cause and cause-specific mortality in a large prospective cohort, the UK Biobank. Of 85,425 participants who had complete information on circulating PUFAs, 6,461 died during follow-up, including 2,794 from cancer and 1,668 from cardiovascular disease (CVD). Associations were estimated by multivariable Cox proportional hazards regression with adjustment for relevant risk factors.

Results

Risk for all three mortality outcomes increased as the ratio of omega-6/omega-3 PUFAs increased (all $P_{\text{trend}} < 0.05$). Comparing the highest to the lowest quintiles, individuals had 26% (95% CI, 15-38%) higher total mortality, 14% (95% CI, 0-31%) higher cancer mortality, and

31% (95% CI, 10-55%) higher CVD mortality. Moreover, omega-3 and omega-6 PUFAs in plasma were all inversely associated with all-cause, cancer, and CVD mortality, with omega-3 showing stronger effects.

Conclusions

Using a population-based cohort in UK Biobank, our study revealed a strong association between the ratio of circulating omega-6/omega-3 PUFAs and the risk of all-cause, cancer, and CVD mortality.

eLife assessment

The manuscript provides **convincing** evidence that both circulating omega-6 and omega-3 PUFAs are associated with lower all-cause, cancer, and cardiovascular mortality in the UK BioBank population and that omega-3s have a stronger effect than omega-6s. The findings have **important** public health implications.

Background

Cancer and cardiovascular disease (CVD) are the two leading causes of non-communicable disease mortality globally.¹ Substantial epidemiologic evidence has linked the dietary or circulating levels of omega-3 and omega-6 polyunsaturated fatty acids (PUFAs) to the risks of all-cause, cancer, and CVD mortality (**Additional file 1: Table S1**). However, the results are inconsistent, especially for omega-6 PUFAs. Most large observational studies support the inverse associations between omega-3 PUFAs and mortality. In addition to total and individual omega-3 PUFAs, the omega-3 index, defined as the percentage of eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) in total fatty acids in red blood cells, was shown to be a validated biomarker of the dietary intake and tissue levels of long-chain omega-3 PUFAs, and was proposed to be a risk factor for CVD and related mortality.^{2,3} While the omega-3 index has been observed to be inversely associated with all cause-mortality, its association patterns with CVD and cancer mortality are less clear.⁴⁻⁶ Most importantly, results from clinical trials of omega-3 PUFA supplementation have been inconsistent.^{7,8} On the other hand, while some studies revealed inverse associations of omega-6 PUFAs with all-cause mortality,⁹⁻¹¹ others reported null results.^{4,12-15} The role of omega-6 PUFAs in cancer and CVD mortality is less studied, and the patterns are similarly conflicting.^{4,9,11} Interpreting previous observational results is challenging due to the limitations of small sample sizes, insufficient adjustments for confounding, and unique sample characteristics. Moreover, many studies rely on self-reported dietary intake or fish oil supplementation status, which are subject to large variability and reporting bias.¹⁶ Despite the tremendous interest and research effort, the roles of omega-3 and omega-6 PUFAs in all-cause and cause-specific mortality remain uncertain.

It has been suggested that the high omega-6/omega-3 ratio in Western diets, 20:1 or even higher, as compared to an estimated 1:1 during the most time of human evolution, contributes to many chronic diseases, including CVD, cancer, and autoimmune disorders.^{17,18} However, while many previous studies have examined total or individual omega-3 and omega-6 PUFAs, fewer investigated the role of their imbalance, as measured by the omega-6/omega-3 ratio, in mortality.^{4,14,15,19} In a prospective cohort study of postmenopausal women, the omega-6/omega-3 ratio in red blood cells was associated with an increased risk of all-cause mortality but

not with cancer or CVD mortality⁴. Similar positive associations with all-cause mortality were observed in smaller cohorts investigating the ratio in serum or dietary intake^{14,19}. However, prospective studies in two independent cohorts from China and US did not find a linear positive association between the omega-6/omega-3 ratio in diet and all-cause mortality¹⁵. To address these gaps in our understanding of the roles of omega-3 and omega-6 PUFAs and their imbalance in all-cause and cause-specific mortality, we perform a prospective study in a large population-based cohort (N = 85,425) from UK Biobank, using objective measurements of PUFA levels in plasma.

Methods

Study population

The UK Biobank study is a prospective, population-based cohort study in the United Kingdom²⁰. Between 2006 and 2010, 502,384 prospective participants, aged 40–69, in 22 assessment Centers throughout the UK were recruited for the study. The population information was collected through a self-completed touch-screen questionnaire; brief computer-assistant interview; physical and functional measures; and blood, urine, and saliva collection during the assessment visit. Participants with cancer (n=37,736) or CVD (n=100,972), those who withdrew from the study (n=879), and those with incomplete data on the plasma omega-6/omega-3 ratio (n=277,372) were excluded from this study, leaving 85,425 participants, 6,461 died during follow-up, including 2,794 from cancer and 1,668 from CVD.

Ascertainment of exposure

Metabolomic profiling of plasma samples was performed with high-throughput nuclear magnetic resonance (NMR) spectroscopy. At the time of this analysis (15 Mar 2023), UK Biobank released the Phase 1 metabolomic dataset, which covered a random selection of 118,461 plasma samples from the baseline recruitment. These samples were collected between 2007 and 2010 and had been stored in -80°C freezers, while the NMR measurements took place between 2019 and 2020.. Detailed descriptions can be found in previous publications about plasma sample preparation, NMR spectroscopy setup, quality control protocols, correction for sample dilution, verification with duplicate samples and internal controls, and comparisons with independent measurements from clinical chemistry assays^{20–22}. Five PUFAs-related biomarkers were directly measured in absolute concentration units (mmol/L), including total PUFAs, total omega-3 PUFAs, total omega-6 PUFAs, docosahexaenoic acid (DHA), and linoleic acid (LA). Of note, DHA is one type of omega-3 PUFAs, and LA is one type of omega-6 PUFAs. Our primary exposure of interest, the omega-6/omega-3 ratio, was calculated based on their absolute concentrations. We also performed supplemental analysis for four exposures, the percentages of omega-3 PUFAs, omega-6 PUFAs, DHA, and LA in total fatty acids (omega-3%, omega-6%, DHA%, and LA%), which were calculated by dividing their absolute concentrations to that of total fatty acids.

Ascertainment of outcome

The date and cause of death were identified through the death registries of the National Health Service (NHS) Information Centre for participants from England and Wales and the NHS Centre Register Scotland for participants from Scotland²⁰. At the time of the analysis (15 Mar 2023), we had access to the most up-to-date mortality dataset (Version: December 2022), which had the date of death up to 12 November 2021. Therefore, follow-up time was calculated as the time between the date of entering the assessment centre and this date, or the date of death, whichever happened first. The underlying cause of death was assigned and coded in vital registries according to the International Classification of Diseases, 10th revision (ICD-10). CVD mortality was defined using codes I00–I99, and cancer mortality was defined using codes C00–D48. To exclude those who had cancer at baseline, we used the overall cancer incidence information based on diagnoses in cancer

registers derived from the Health and Social Care Information Centre and the NHS Central Register that were coded based on ICD-9 and ICD-10 codes. The information about CVD incidence was retrieved from Hospital inpatient data and Death Register records that were coded according to ICD-9 and ICD-10 codes, and self-reported medical condition.

Ascertainment of covariates

The baseline questionnaire included detailed information on several possible confounding variables: demographic factors (age, sex, assessment centre, ethnicity), socioeconomic status (Townsend Deprivation Index), lifestyle habits (alcohol assumption, smoking status, body mass index (BMI), physical activity, comorbidities (including hypertension, diabetes, and longstanding illness)), and other supplementation (Fish-oil supplementation). The Townsend Deprivation Index, used as an indicator of socioeconomic status, is retrieved directly from the UK Biobank. BMI was defined as the body mass divided by the square of the body height and was expressed in units of kg/m^2 . Comorbidities, including hypertension, diabetes, and longstanding illness, were self-reported at baseline. Longstanding illness refers to any long-standing illness, disability, or infirmity, without other specific information. We also retrieved the information on dietary PUFA intakes and serum biomarkers for secondary analysis. Dietary PUFAs were estimated based on the 24-hour dietary recall. The percentage of dietary omega-3 and omega-6 PUFA were calculated as the corresponding absolute dietary PUFA divided by summation of dietary monounsaturated fatty acids, saturated fatty acids, omega-3 fatty acids, and omega-6 fatty acids. The biochemistry markers were measured in the blood sample collected at recruitment.

Statistical analysis

We summarized and compared the characteristics of the participants across quintiles of the omega-6/omega-3 ratio at baseline using descriptive statistics. Pearson's Chi-squared test and ANOVA test were used to compare the demographic characteristics across quintiles, respectively, for categorical variables and continuous variables. To investigate associations of the ratio with all-cause and cause-specific mortality, we used multivariable Cox proportional hazards regression models to calculate hazard ratios and their 95% confidence intervals (CI). The proportional hazards assumption was not violated based on Schoenfeld residuals. We analyzed the ratio as continuous and categorical variables (i.e., quintiles). For all trend tests, we used the median value of each quintile as a continuous variable in the models. Potential nonlinear associations were assessed semi-parametrically using restricted cubic splines (4 knots were used in regression splines).

Based on previous literature and biological plausibility, we chose the following variables as covariates in the multivariate models: age (years; continuous), sex (male, female), race (White, Black, Asian, Others), Townsend deprivation index (continuous), assessment centre, BMI (kg/m^2 ; continuous), smoking status (never, previous, current), alcohol intake status (never, previous, current), physical activity (low, moderate, high), and comorbidities (yes, no).

In secondary analyses to assess potential differences in associations across different population subgroups, we repeated the above-described analyses stratified by age (< vs. $\geq$ the median age of 58 years), sex (male/female), Townsend deprivation index (< vs. $\geq$ the population median of -2), BMI (< vs. $\geq$ 25), comorbidities (yes vs. no), physical activity (low and moderate vs. high) and current smoking status (yes vs. no). Besides, to compare the effects of dietary intake of PUFAs with the plasma levels on mortality, we repeated the above-described analyses in participants with complete information on dietary intake PUFAs (N=153,064). Moreover, we conducted mediation analysis to investigate potential biological pathways that might explain the effects of omega-6/omega-3 ratio on mortality outcomes. Based on plausibility, we chose CRP as a potential mediating factor for CVD mortality; SHBG, TTST, E2 and IGF-1 as potential mediating factors for cancer mortality; and all the above biomarkers for all-cause mortality. We first examined the variations in the concentrations of each biomarker according to omega-6/omega-3 ratio quintiles.

Then we constructed two regression models, a linear model, to regress the mediator (treated the biomarkers as continuous) on the exposure (treated the omega-6/omega-3 ratio as continuous), with fully-adjustment for potential confounders; and a Cox proportional hazard model, to regress the outcome (mortality outcomes) on the exposure and the mediator, with fully-adjustment for potential confounders. We chose to construct Cox models for mediation analysis because our outcomes were rare (all <10%)²⁸. We integrated these two regression models to obtain the indirect effects and proportions of mediating effects through the counterfactual-based approach proposed by Vander Weele^{29,30}.

We also conducted several sensitivity analyses. First, to assess whether the associations of the omega-6/omega-3 ratio with mortality outcomes were primarily driven by omega-3 fatty acids or omega-6 fatty acids, we assessed both the separate and the joint associations of omega-3 fatty acids to total fatty acids percentage (omega-3%) and omega-6 fatty acids to total fatty acids percentage (omega-6%) with the three mortality outcomes. We also performed a joint analysis with categories of the omega-3% and omega-6% quintiles, using participants in both the lowest omega-3% and omega-6% quintiles as the reference category. An interaction term between omega-3% and omega-6% was included in the multivariable Cox proportional hazards model, and a Likelihood Ratio test was used to assess its significance. The correlation between omega-3% and omega-6% was assessed by the Pearson correlation. Second, to assess the effects of individual PUFAs on mortality, we conducted the same analysis for circulating docosahexaenoic acid to total fatty acids percentage (DHA%) and linoleic acid to total fatty acids percentage (LA%) with the three mortality outcomes. Third, to address the potential residual confounding by fish oil supplementation, we further adjusted for the supplementation status from the baseline questionnaire. Fourth, to investigate the effects of missing values, we imputed missing values (<1% for most factors, up to 19% for physical activity) by chained equations and performed synthesis analyses on the imputed datasets³¹. Fifth, to assess whether the observed associations are attenuated by reverse causation, we excluded those who died in the first year of follow-up. Last, to assess the representativeness of the participants included in our study, we compared the baseline characteristics between the participants with and without exposure information. All P-values were two-sided. We considered a P-value <0.05 or a 95% confidence interval (CI) excluding 1.0 for HRs as statistically significant. We conducted all analyses using R, version 4.0.3 and SAS statistical software, version 9.4 (SAS Institute Inc.).

Results

Baseline characteristics

In the analytic cohort of 85,425 participants, over a mean of 12.7 years of follow-up, 6,461 died, including 2,794 from cancer and 1,668 from CVD. The baseline characteristics of the participants across quintiles of the ratio of omega-6/omega-3 were summarized in **Table 1**. Study participants were, on average, 56 years old and 90% White. Those in the higher ratio quintiles were more likely to be younger, male, and current smokers, but less likely to have comorbidities and take fish oil supplementation.

Main results

The associations of the omega-6/omega-3 ratio with all-cause and cause-specific mortality risks were presented in **Table 2**. A higher ratio was strongly associated with higher mortality from all causes, cancer, and CVD ($P_{\text{trend}} < 0.05$ for all three). In the fully adjusted models that considered the ratio as a continuous variable, every unit increase in the ratio corresponded to 2%, 1%, and 2% higher risk in all-cause, cancer, and CVD mortality, respectively. When comparisons were made between the highest and the lowest quintile of the omega-6/omega-3 ratio, there were 26%, 14%, and 31% increased risk for all-cause, cancer, and CVD mortality, respectively.

Characteristics ^a	1 (median = 5.9) (n = 17,085)	2 (median = 7.6) (n = 17,085)	3 (median = 9.1) (n = 17,085)	4 (median = 11.0) (n = 17,085)	5 (median = 14.8) (n = 17,085)	P
Age (years)	58.6 (7.5)	57.2 (7.9)	55.7 (8.2)	54.7 (8.3)	53.4 (8.2)	<0.001 ^b
Sex (male%)	40.3	44.9	47.5	49.4	52.8	<0.001 ^c
Ethnicity (n%)						
White	15,375 (90.4%)	15,486 (91.0%)	15,494 (91.1%)	15,467 (91.1%)	15,420 (90.7%)	0.384 ^c
Black	113 (0.7%)	99 (0.6%)	105 (0.6%)	92 (0.5%)	116 (0.7%)	
Asian	673 (4.0%)	639 (3.8%)	663 (3.9%)	653 (3.8%)	659 (3.9%)	
Others	844 (5.0%)	786 (4.6%)	742 (4.4%)	773 (4.6%)	810 (4.8%)	
Missing (n)	80	75	81	100	80	
BMI	27.0 (4.4)	27.3 (4.5)	27.3 (4.7)	27.3 (4.8)	26.9 (5.0)	<0.001 ^b
Missing (n)	54	64	55	58	71	
TDI	-1.6 (3.0)	-1.5 (3.0)	-1.4 (3.1)	-1.2 (3.1)	-0.9 (3.2)	<0.001 ^b
Missing (n)	17	24	24	28	22	
Smoking status (n%)						<0.001 ^c
Never	9,426 (55.5%)	9,272 (54.6%)	9,424 (55.4%)	9,400 (55.3%)	9,214 (54.2%)	
Previous	6,402 (37.7%)	6,194 (36.5%)	5,772 (34.0%)	5,492 (32.3%)	4,951 (29.1%)	
Current	1,162 (6.8%)	1,526 (9.0%)	1,802 (10.6%)	2,104 (12.4%)	2,832 (16.7%)	
Missing (n)	95	93	87	89	88	
Alcohol status (n%)						<0.001 ^c
Never	631 (3.7%)	676 (4.0%)	655 (3.8%)	755 (4.4%)	974 (5.7%)	
Previous	536 (3.1%)	554 (3.2%)	535 (3.1%)	617 (3.6%)	835 (4.9%)	
Current	15,877 (93.2%)	15,822 (92.8%)	15,847 (93.0%)	15,664 (91.9%)	15,221 (89.4%)	
Missing (n)	41	33	48	49	55	
Physical activity (n%)						<0.001 ^c
Low	2,422 (17.4%)	2,602 (18.8%)	2,653 (19.1%)	2,617 (19.0%)	2,613 (18.9%)	
Moderate	5,879 (42.2%)	5,762 (41.5%)	5,712 (41.1%)	5,533 (40.1%)	5,327 (38.6%)	
High	5,635 (40.4%)	5,506 (39.7%)	5,545 (39.9%)	5,654 (41.0%)	5,854 (42.4%)	
Missing (n)	3,149	3,215	3,175	3,281	3,291	
Fish oil supplementation (Yes%)	48.1	37.7	29.6	22.7	15.5	<0.001 ^c ¹
Missing (n)	50	47	77	82	71	
Comorbidity (Yes%)	38.6	37.3	34.2	33.8	33.0	<0.001 ^c
Plasma omega-3 percentage	6.6 (1.4)	4.9 (0.5)	4.2 (0.4)	3.5 (0.3)	2.6 (0.5)	<0.001 ^b
Plasma omega-6 percentage	36.3 (3.7)	37.4 (3.4)	38.2 (3.3)	39.1 (3.2)	40.2 (3.1)	<0.001 ^b
Plasma DHA percentage	2.8 (0.8)	2.1 (0.5)	1.9 (0.4)	1.7 (0.4)	1.5 (0.4)	<0.001 ^b
Plasma LA percentage	27.0 (3.3)	28.4 (3.1)	29.3 (3.0)	30.2 (3.0)	31.3 (3.1)	<0.001 ^b
Dietary omega-3 percentage	3.6 (1.8)	3.2 (1.4)	3.0 (1.3)	2.9 (1.2)	2.7 (1.0)	<0.001 ^b
Missing (n)	9,355	9,628	9,857	9,974	10,288	
Dietary omega-6 percentage	16.8 (4.6)	16.5 (4.4)	16.6 (4.6)	16.5 (4.7)	16.5 (4.8)	<0.001 ^b
Missing (n)	9,355	9,628	9,857	9,974	10,288	

^a All variables measured at baseline are presented as mean (SD) unless otherwise specified.

^b From the ANOVA test for continuous variables.

^c From the Pearson's Chi-squared test for categorical variables.

Table 1.

Selected participants characteristics at baseline across quintiles of the plasma omega-6/omega-3 PUFAs ratio (n=85,425).

Omega ratio variable forms	Causes of death								
	All-cause			Cancer			Cardiovascular diseases		
	Number of deaths	Partially adjusted associations ^b	Fully adjusted associations ^c	Number of deaths	Partially adjusted associations ^b	Fully adjusted associations ^c	Number of deaths	Partially adjusted associations ^b	Fully adjusted associations ^c
		HR (95% CI)	HR (95% CI)		HR (95% CI)	HR (95% CI)		HR (95% CI)	HR (95% CI)
Continuous	6,461	1.02 (1.02-1.03)	1.02 (1.02-1.03)	2,794	1.02 (1.01-1.03)	1.01 (1.00-1.02)	1,668	1.02 (1.01-1.03)	1.02 (1.01-1.03)
Quintiles (median)									
1 (5.9)	1,348	1.00 (ref)	1.00 (ref)	593	1.00 (ref)	1.00 (ref)	369	1.00 (ref)	1.00 (ref)
2 (7.6)	1,256	1.00 (0.92-1.08)	0.96 (0.88-1.05)	563	1.02 (0.91-1.15)	0.98 (0.86-1.12)	315	0.90 (0.77-1.05)	0.89 (0.75-1.06)
3 (9.1)	1,236	1.06 (0.98-1.15)	1.01 (0.93-1.11)	543	1.08 (0.96-1.21)	0.99 (0.87-1.13)	321	0.97 (0.84-1.13)	0.97 (0.81-1.16)
4 (11.0)	1,252	1.14 (1.06-1.23)	1.09 (1.00-1.19)	548	1.16 (1.03-1.31)	1.12 (0.98-1.27)	306	0.98 (0.84-1.15)	1.02 (0.86-1.22)
5 (14.8)	1,369	1.34 (1.24-1.45)	1.26 (1.15-1.38)	547	1.27 (1.12-1.43)	1.14 (1.00-1.31)	357	1.20 (1.04-1.40)	1.31 (1.10-1.55)
<i>P</i> _{trend}		<0.001	<0.001		<0.001	0.011		0.002	<0.001

Abbreviations: CI, confidence interval; HR, hazards ratio; ref, reference.

^a From Cox proportional hazards regression.

^b Adjusted for age (years; continuous), sex (male, female), race (White, Black, Asian, Others), Townsend deprivation index (continuous), assessment centre.

^c Adjusted for age (years; continuous), sex (male, female), race (White, Black, Asian, Others), Townsend deprivation index (continuous), assessment centre, BMI (kg/m²; continuous), smoking status (never, previous, current), alcohol intake status (never, previous, current), physical activity (low, moderate, high), and comorbidities (yes, no).

Table 2.

Associations^a of the plasma omega-6/omega-3 PUFAs ratio with all-cause, cancer, and CVD mortality risk in the UK Biobank.

Stratified analysis

The fully adjusted associations of the omega-6/omega-3 ratio with all-cause mortality revealed that compared to the lowest quintile, the highest quintile has strong, statistically significant associations with elevated risk within all categories of age, sex, TDI, BMI, comorbidities, physical activity, and smoking status (**Figure 1**, **Additional file 2: Table S2**), except in those aged less than 58 years old and BMI less than 25. The estimated associations with all-cause mortality were stronger in current smokers (P for interaction < 0.01 ; **Figure 1**, **Additional file 2: Table S2**). For cancer and CVD mortality, they also tended to be stronger among current smokers but did not reach statistical significance (P for interaction = 0.13 and 0.12, respectively). The associations with CVD mortality tended to be stronger among participants without comorbidities (P for interaction < 0.01). No significant interactions were found for other risk factors (**Figure 2**).

Restricted cubic spline analysis

Restricted cubic spline analysis suggested significant positive associations of the omega-6/omega-3 ratio with all-cause, cancer, and CVD mortality ($P < 0.05$ for all three outcomes, **Figure 3**). Potential nonlinearity in these positive associations was identified for all-cause mortality ($P < 0.05$) and CVD mortality ($P < 0.05$) but not for cancer mortality ($P = 0.12$). The strength of the relationship between the ratio and all-cause mortality appears to remain at a relatively low level before it starts to increase quickly after the ratio exceeds 8. A similar trend with higher uncertainties was observed for CVD mortality.

Omega-3, Omega-6, Joint analysis, and comparison with dietary PUFAs

We further performed analyses to assess whether the associations of the omega-6/omega-3 ratio with mortality outcomes were primarily driven by omega-3 or omega-6 fatty acids. The correlation between omega-3% and omega-6% was relatively low with $r = -0.12$ ($P < 0.01$). Across all models, both the omega-3% and omega-6% were inversely associated with all three mortality outcomes, except for plasma omega-6% with CVD mortality under Model 3 ($P_{\text{trend}} < 0.01$, **Additional file 2: Table S3 and Table S4**). Notably, their associations remained significant when they were included in the same models. On the other hand, the effect sizes of the inverse associations were always bigger for the omega-3% under the fully adjusted Model 3. For example, when comparing those in the highest omega-3% quintile to the lowest quintile, the fully adjusted HRs (95% CI) for all-cause, cancer, and CVD mortality were, respectively, 0.69 (0.63, 0.76), 0.75 (0.65, 0.87), and 0.68 (0.57, 0.82) (**Additional file 2: Table S3**). The corresponding HRs for the omega-6% were 0.77 (0.70, 0.85), 0.80 (0.68, 0.92), and 0.83 (0.68, 1.02) (**Additional file 2: Table S4**). Furthermore, in another joint analysis of the omega-3% and omega-6%, the lowest risk for all-cause and cancer mortality was observed among those in the joint highest categories of the two fatty acids (**Additional file 2: Table S5**). For example, when comparing those in the highest quintiles of the two fatty acids to the group with the joint lowest group, the HRs (95% CI) for all-cause and cancer mortality were, respectively, 0.48 (95% CI, 0.35, 0.67) and 0.53 (95% CI, 0.33, 0.86). In the analysis of dietary PUFAs with mortality, the effect sizes were smaller and less significant compared with those of the corresponding plasma levels. When analyzed as continuous variables, dietary omega-3%, omega-6% and omega-6/omega-3 ratio were all significantly associated with all-cause, cancer, and CVD mortality, in directions consistent with their plasma counterparts. The trend test across the five quintiles revealed significant associations between dietary omega-3% and omega-6/omega-3 ratio with cancer mortality (P for trend < 0.001 and $= 0.002$, respectively; **Additional file 2: Table S6**). The correlation between dietary omega-3% and omega-6% was $r = 0.41$ ($P < 0.01$).

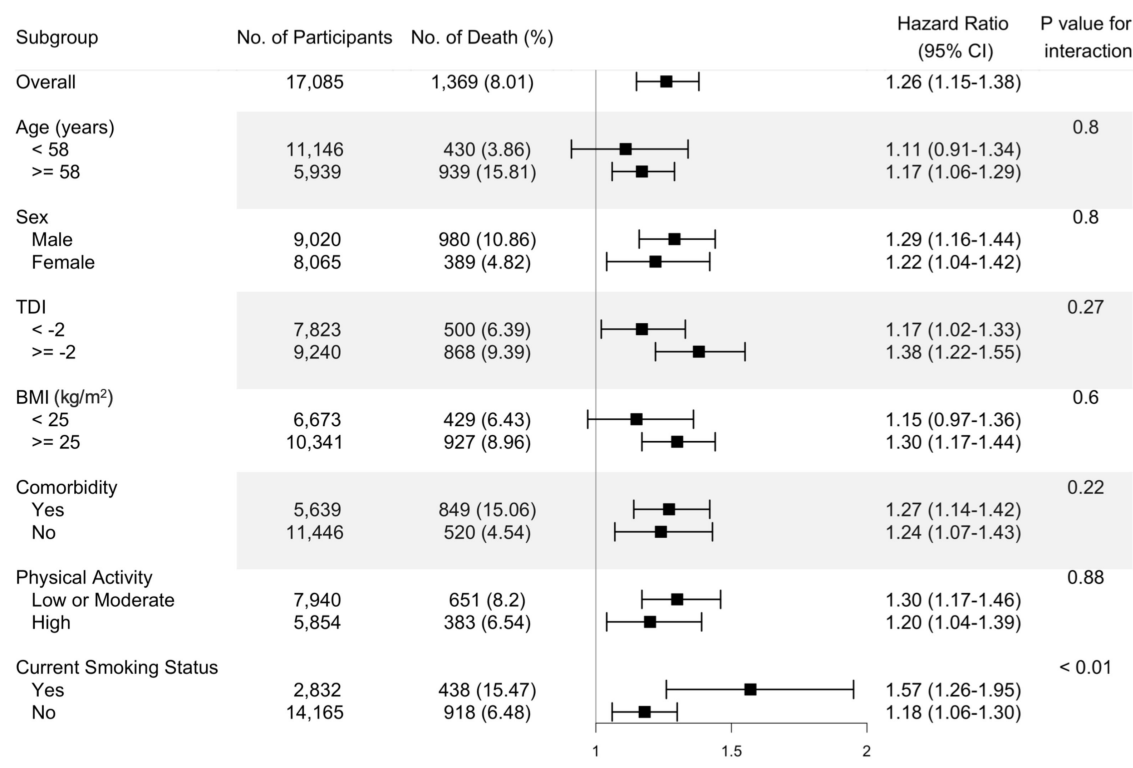


Figure 1.

Risk estimates of all-cause mortality for the highest compared with the lowest quintile of the ratio of plasma omega-6 to omega-3 PUFAs, stratified by potential risk factors. Results were adjusted for age (years; continuous), sex (male, female), race (White, Black, Asian, Others), Townsend deprivation index (continuous), assessment centre, BMI (kg/m²; continuous), smoking status (never, previous, current), alcohol intake status (never, previous, current), physical activity (low, moderate, high), and comorbidities (yes, no).

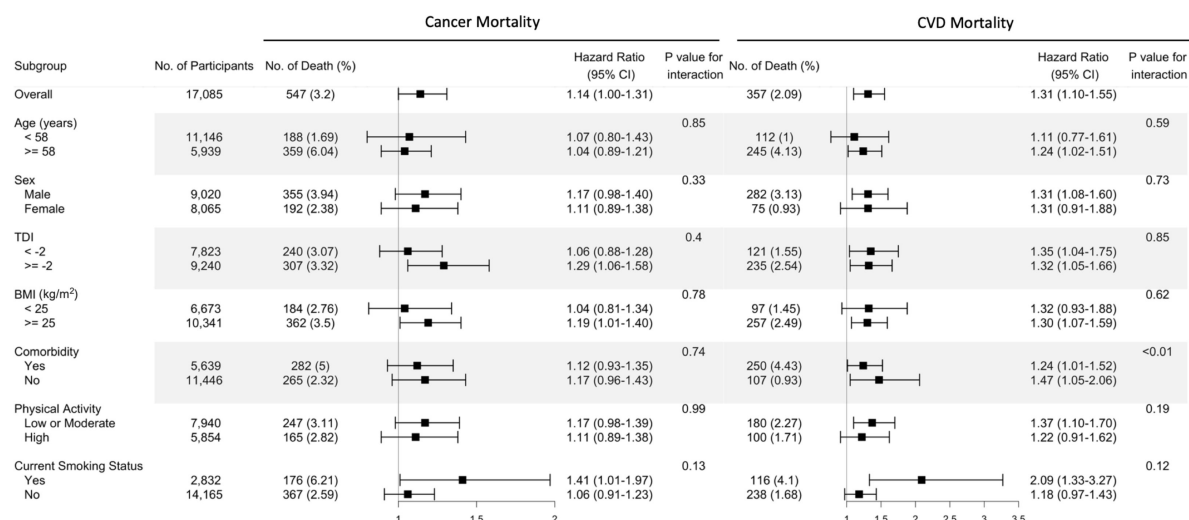


Figure 2.

Risk estimates of cause-specific mortality for the highest compared with the lowest quintile of the ratio of plasma omega-6 to omega-3 PUFAs, stratified by potential risk factors. Results were adjusted for age (years; continuous), sex (male, female), race (White, Black, Asian, Others), Townsend deprivation index (continuous), assessment centre, BMI (kg/m²; continuous), smoking status (never, previous, current), alcohol intake status (never, previous, current), physical activity (low, moderate, high), and comorbidities (yes, no).

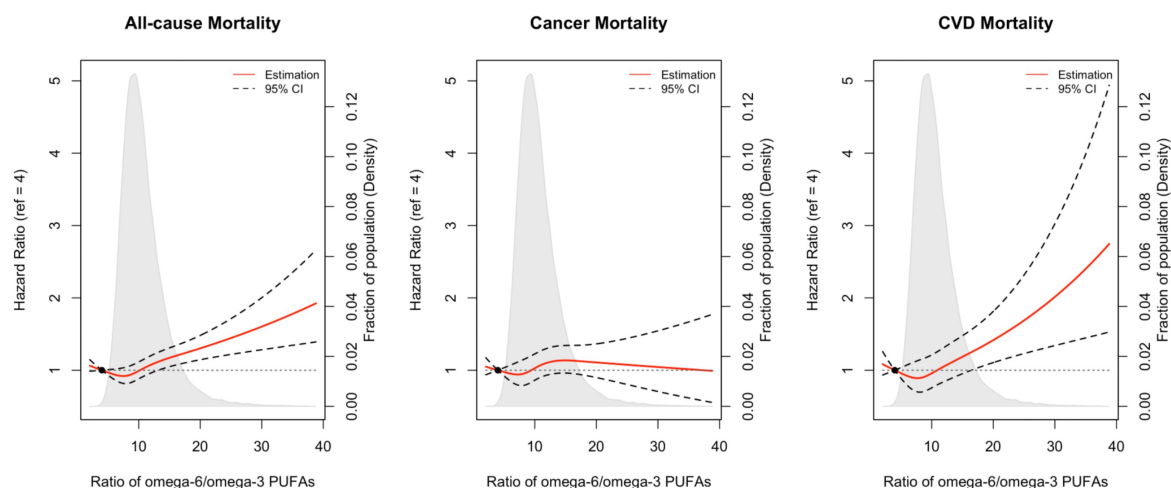


Figure 3.

Associations of the ratio of omega-6/omega-3 PUFAs with all-cause and cause-specific mortality evaluated using restricted cubic splines. Hazard ratios and omega ratios are presented in the vertical and horizontal axis, respectively.

The best estimates and their confidence intervals are presented as solid red lines and dotted black lines, respectively. The ratio 4 was selected as a reference level, and the x-axis depicts the ratio from 0 to 40. Potential nonlinearity was identified for all-cause mortality ($p < 0.05$) and CVD-caused mortality ($p < 0.05$), but not for cancer-caused mortality ($p = 0.12$). All HRs are adjusted for age (years; continuous), sex (male, female), race (White, Black, Asian, Others), Townsend deprivation index (continuous), assessment centre, BMI (kg/m²; continuous), smoking status (never, previous, current), alcohol intake status (never, previous, current), physical activity (low, moderate, high), and comorbidities (yes, no).

Mediation analysis for intermediate biomarkers

The baseline serum biomarkers of the participants across quintiles of the ratio of omega-6/omega-3 were summarized in **Additional file 2: Table S7**. The estimates for the three paths a (association between plasma omega-6/omega-3 ratio and biomarkers), b (association between biomarkers and mortality outcomes), and c (association between plasma omega-6/omega-3 ratio and mortality outcomes), as well as the percentage of indirect effect (a to b) among the overall associations, were reported in **Additional file 2: Table S8 and Figure S1**. These analyses identified CRP and SHBG as potential mediators for the association of plasma omega-6/omega-3 ratio with all-cause mortality (proportion of mediating = 4% and 5.0%, respectively).

Other sensitivity analyses

We performed analyses to examine the associations of DHA% and LA% with mortalities. The correlation between DHA% and LA% was relatively low with $r = 0.03$ ($P < 0.01$). Under the fully-adjusted model, both DHA% and LA% were inversely associated with all three mortality outcomes, regardless of treated as continuous or quintiles (**Additional file 2: Table S9**). After further adjustment of the fish oil supplementation status, the associations between the ratio of omega-6/omega-3 and mortality outcomes were slightly attenuated yet did not alter the main findings (**Additional file 2: Table S10**). Our primary analysis excluded participants with missing information (i.e., a complete-case analysis). We performed a sensitivity analysis using the multiply-imputed datasets, and there were no substantial changes (**Additional file 2: Table S11**). Moreover, the exclusion of participants who died during their first-year follow-up did not materially alter the results (**Additional file 2: Table S12**). The baseline characteristics are comparable between participants with or without exposure information (**Additional file 2: Table S13**).

Discussion

In this prospective population-based study of UK individuals, we showed that a higher ratio of plasma omega-6/omega-3 fatty acids was positively associated with the risk of all-cause, cancer, and CVD mortality. These associations were independent of most risk factors examined, including age, sex, TDI, BMI, comorbidities, and physical activity, but they were stronger in current smokers. These relationships were linear for cancer mortality but not for all-cause and CVD mortality. For those two outcomes, the risk of mortality first decreased at lower ratios and then increased, with an inflection point around the ratio of 8. Moreover, omega-3 and omega-6 PUFAs in plasma were consistently and inversely associated with all-cause, cancer, and CVD mortality, with omega-3 showing stronger effects.

To date, studies that examined the relationship between the ratio of omega-6/omega-3 PUFAs and mortality in the general population are sparse^{4,14,15,19}. Similar to our finding, in a 2017 report from a prospective women cohort study ($n=6,501$; 1,875 all-cause deaths, 617 CVD deaths, 462 cancer deaths)⁴, the adjusted HR for all-cause mortality was 1.10 (1.02-1.19) per 1-SD increase of omega-6/omega-3 ratio in red blood cells, however, the effects were not significant for cancer and CVD deaths. Another study supporting our results was conducted on elderly Japanese individuals ($n=1,054$; 422 deaths). It found that the ratio of an omega-3 fatty acid, eicosapentaenoic acid, to an omega-6 fatty acid, arachidonic acid (ARA), is inversely associated with all-cause mortality, with an HR of 0.71 (95% 0.53–0.96) comparing the highest to the lowest tertile¹⁴. Findings of previous studies based on dietary intake were null^{15,19}. A prospective cohort involving 145 hemodialysis patients enrolled in Southern California during 2001-2007 (42 all-cause deaths) showed that the estimated HR (95% CI) for all-cause mortality among those in the lowest relative to the highest quartiles of dietary omega-6/omega-3 ratio was 0.37 (0.14-1.08)¹⁹. Although the effect was not significant, it still shows the protective trend of a lower omega-6/omega-3 ratio against premature death, when taking the sample size and study population into consideration. In a 2019 report based on two population-based prospective cohorts in China

($n=14,117$, 1,007 all-cause deaths) and the US ($n=36,032$, 4,826 all-cause deaths), the effect of the omega-6/omega-3 ratio intake was not significant (HR (95% CI) is 0.95 (0.80-1.14) for China and 0.99 (0.89-1.11) for US, respectively)¹⁵. These discrepancies may be explained by the usage of circulating biomarkers or dietary intakes for calculating the omega-6/omega-3 ratio. Indeed, we performed the same association analyses for dietary PUFAs and found that their associations with mortality have smaller effect sizes and are less statistically significant, although maintaining the same effect directions as plasma PUFAs. The estimated dietary intakes may be inaccurate due to recall bias or outdated food databases¹⁶. The circulating level is a more objective measurement of PUFA status and thus provides a more reliable picture of the effects of omega-3 and omega-6 PUFAs on mortality. Additionally, we observed that the omega-6/omega-3 ratio tends to have stronger effects on mortality outcomes in current smokers. This is consistent with a previous observation that the association between habitual fish oil supplementation and a lower risk of all-cause mortality is stronger in current smokers²⁶. Moreover, we found that current smokers in UK Biobank tend to have higher omega-6/omega-3 ratios. Similarly, prior studies reported that current smokers have lower circulating omega-3 PUFAs than non-smokers but similar circulating levels of omega-6 PUFAs^{32,33}. It is likely that omega-3 PUFAs play more important roles in current smokers, resulting in a stronger effect of the omega-6/omega-3 ratio on mortality. However, the underlying mechanism awaits future investigations.

A large number of existing observational studies documented the inverse association of circulating levels and intake of omega-3 PUFAs with mortality^{5,6,27,34,39}, which is in line with our finding that individuals in the highest quintile of the plasma omega-3% had approximately 30% lower risk for all-cause and CVD mortality and 25% lower risk for cancer mortality, when compared to those in the lowest quintile. Our analysis of dietary omega-3% also revealed inverse associations with all-cause and cancer mortality, with 9% and 18% lower risk, respectively. Few previous studies have examined the association in generally healthy populations^{5,27,34,36}. In a 2016 meta-analysis³⁴ of seven epidemiologic studies for dietary omega-3 PUFAs intake and four studies for circulating levels, the estimated relative risk for all-cause mortality was 0.91 (95% CI: 0.84-0.98) when comparing the highest to the lowest categories of omega-3 PUFA intake. In two 2018 reports of population-based studies in the US, the circulating level and dietary intake of omega-3 PUFAs were significantly associated with lower total mortality^{5,27}. In a 2021 meta-analysis of 17 epidemiologic studies, comparing the highest to the lowest quintiles of circulating EPA and DHA, the estimated HRs for all-cause, CVD, and cancer mortality were 0.84 (0.79-0.89), 0.80 (0.73-0.88) and 0.87 (0.78-0.96), respectively³⁶. Moreover, inverse associations of omega-3 PUFAs biomarkers with total mortality were found in patients with myocardial infarction^{37,39}, type 2 diabetes¹², and other diseases^{6,35,38,40}; inverse associations of omega-3 PUFAs with CVD mortality were also reported in these patient groups^{6,12,39}. However, there are also reports of a null relationship between omega-3 PUFAs and mortality^{13,14,41}. One possible explanation for the discrepancies could be the limited statistical power due to the small sample size and the small number of events. Moreover, the inconsistency may be due to unique sample characteristics; one study only involved elder patients¹⁴, and one only involved women⁴¹.

A smaller number of studies have evaluated the associations of omega-6 PUFAs with mortality. Our findings in the large UK prospective cohort are in accordance with the results of several previous studies that increased circulating levels of omega-6 PUFAs were associated with decreased all-cause mortality^{9,11,15,42}. Moreover, a 2019 meta-analysis involving 18 cohort studies and 12 case-control studies showed that an omega-6 PUFA, linoleic acid, was inversely associated with CVD mortality; the HR per interquintile range was 0.78 (95% CI, 0.70-0.85)¹⁰. Other studies, however, did not support such inverse associations^{4,12,14,41}. Although the findings were inconsistent, no previous studies reported harmful effects of omega-6 PUFAs on mortality^{4,11,15,41-44}. Our studies support the inverse associations between omega-6 PUFAs and all-cause and cause-specific mortality. Further research is needed to investigate the health impacts of omega-6 PUFAs in laboratory studies, epidemiological investigations, and clinical trials.

We observed that the omega-6/omega-3 ratio is positively, while both of its numerator and denominator, omega-6% and omega-3%, are negatively associated with mortality. Our findings support that both omega-6 and omega-3 PUFAs are protective against death and that the positive associations of the omega-6/omega-3 ratio with mortality outcomes are likely due to the stronger effects of omega-3 than omega-6 PUFAs. Our mediation analysis with clinical biomarkers for CVD and cancer revealed that CRP and SHBG partially mediated the effects of the omega-6/omega-3 ratio on all-cause mortality. However, the proportions mediated are relatively small (~5%), suggesting the presence of direct effects or unexamined factors. Our findings call for future studies that evaluate the utility and investigate the mechanisms of the omega-6/omega-3 ratio.

Strengths of our current study include the use of objective PUFA biomarkers in plasma instead of the estimated intakes from dietary questionnaires, which increases the accuracy of exposure assessment. Moreover, the prospective population-based study design, large sample size, long duration of follow-up, and detailed information on potential confounding variables, substantially mitigate the possible complications from reverse causality and confounding bias. In several sensitivity analyses, most of the documented associations remain materially unchanged, indicating the robustness of our results.

Several potential limitations deserve attention. First, plasma omega-3 and omega-6 PUFAs were measured only once at baseline. Their levels may vary with diet or other lifestyle factors, which could cause misclassification over follow-up. However, some studies demonstrated that multiple measurements of omega-3 PUFAs have been consistent for a 6-month period⁴⁵. Moreover, the 13-year within-person correlation for circulating omega-6 PUFAs was comparable to such correlations for other major CVD risk factors⁴⁶. Thus, the single measurement of PUFAs at baseline, although not perfect, provides us with adequate information to investigate the relative long-term effects. Second, although we adjusted for many potential confounders in the model, we cannot rule out the imprecisely measured and unmeasured factors. Third, we acknowledged that the cohort did not mirror the broader UK demographic in terms of socioeconomic and health profiles. Participants in the UK Biobank generally exhibited better health and higher socioeconomic status than the average UK resident, potentially influencing the disease prevalence and incidence rates. Nonetheless, the UK Biobank's extensive sample size and comprehensive exposure data enable the generation of valuable estimates for exposure-disease associations. These estimates have been corroborated by findings from more demographically representative cohorts^{47,48}. Last, although we included individuals of different ancestries in the analysis, over 90% of the participants were of European ancestry. The generalizability of our findings across ancestries requires future verification.

Conclusions

In this large prospective cohort study, we documented robust positive associations of the plasma omega-6/omega-3 fatty acids ratio with the risk of all-cause, cancer, and CVD mortality. Moreover, we found that plasma omega-3 and omega-6 PUFAs were independently and inversely associated with the three mortality outcomes, with omega-3 fatty acids showing stronger inverse associations. Our findings support the active management of a high circulating level of omega-3 fatty acids and a low omega-6/omega-3 ratio to prevent premature death. Future research is warranted to further test the causality, such as Mendelian randomization and randomized controlled trials. Mechanistic research, including comprehensive mediation analysis, in-depth experimental characterization in animal models or cell lines, and intervention studies, is also needed to unravel the molecular and physiological underpinnings.

Declarations

Ethics approval and consent to participate

The UK Biobank received ethical approval from the research ethics committee (reference ID: 11/NW/0382). Written informed consent was obtained from participants.

Consent for publication

Not applicable.

Availability of data and materials

The datasets analyzed during the current study are available from the UK Biobank through an application process (www.ukbiobank.ac.uk/ )

Competing interests

The authors declare no competing interests.

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Authors' contributions

YZ performed data analysis, prepared visualizations, and wrote the original draft of the manuscript. YS, QY, and SS contributed to the data analysis. YS (yeshen@uga.edu) and KY (kaixiong.ye@uga.edu) contributed equally to this project and should be considered co-corresponding authors. They jointly designed and supervised the project. JTB contributed to the interpretations of results. All authors critically edited the manuscript for important intellectual content. The corresponding author (KY) attests that all listed authors meet the authorship criteria and that no others meeting the criteria were omitted.

Data Availability

The datasets analyzed during the current study are available from the UK Biobank through an application process (www.ukbiobank.ac.uk/ )

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Abbreviations

- 95%: CI 95% confidence interval
- ANOVA: Analysis of variance
- BMI: Body mass index
- CRP: C-creative protein

- CVD: Cardiovascular disease
- DHA: Docosahexaenoic acid
- DHA%: Docosahexaenoic acid to total fatty acids percentage
- EPA: Eicosapentaenoic acid
- HR: Hazard ratio
- LA: Linoleic acid
- LA%: Linoleic acid to total fatty acids percentage
- IGF-1: Insulin-like growth factor 1
- Omega-3%: Omega-3 fatty acids to total fatty acids percentage
- Omega-6%: Omega-6 fatty acids to total fatty acids percentage
- PUFAs: Polyunsaturated fatty acids
- Ref: Reference
- SD: Standard deviation
- SHBG: Sex hormone binding globulin
- TTST: Testosterone

Supplementary Information

Additional file 1: Table S1. Literature review.

Additional file 2: Table S2. Risk estimates of ratio of plasma omega-6 to omega-3 PUFAs with all-cause, cancer and CVD mortality, stratified by potential risk factors, in the UK Biobank Study.

Table S3. Associations of plasma omega-3 PUFAs percentage with all-cause, cancer, and CVD mortality risk in the UK Biobank Study. **Table S4.** Associations of plasma omega-6 PUFAs percentage with all-cause, cancer, and CVD mortality risk in the UK Biobank Study. **Table S5.** Fully adjusted joint associations of plasma omega-3 PUFAs percentage and omega-6 PUFAs percentage with all-cause and cause-specific mortality in the UK Biobank Study. **Table S6.** Associations of the dietary PUFAs with all-cause, cancer, and CVD mortality risk in the UK Biobank. **Table S7.** Selected participants serum biochemical markers at baseline across quintiles of the plasma omega-6/omega-3 PUFAs ratio. **Table S8.** Mediation analysis of biomarkers on associations between plasma omega-6/omega-3 ratio and mortality. **Figure S1.** Directed acyclic graph to explain mediation. **Table S9.** Associations of DHA and LA with all-cause, cancer, and CVD mortality risk in the UK Biobank. **Table S10.** Associations of ratio of omega-6/omega-3 PUFAs with all-cause, cancer, and CVD mortality risk in the UK Biobank, covariates including fish oil supplementation status. **Table S11.** Associations of ratio of omega-6/omega-3 PUFAs with all-cause, cancer, and CVD mortality risk in the UK Biobank Study, with multiple imputation for missing data. **Table S12.** Associations of ratio of omega-6/omega-3 PUFAs with all-cause, cancer, and CVD mortality risk in the UK Biobank Study, excluding those who died in the first follow-up year. **Table S13.** Baseline characteristics of participants with missing exposure information and not included in the study.

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Editors

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Reviewer #3 (Public Review)

Summary:

The authors are trying to find out whether the levels of omega-6 and omega-3 fatty acids in the blood are linked to the likelihood of dying from anything, of dying from cancer and of dying from cardiovascular disease. They use a large dataset called UKBiobank where fatty acid levels were measured in blood at the start of the study and what happened to the participants over the following years (average of 12.7 years) was followed. They find that both omega-6 AND omega-3 fatty acids were linked with less likelihood of dying from anything, from cancer and from cardiovascular disease. The effects of omega-3s were stronger. They then made a ratio of omega-6 to omega-3 fatty acids and found that as that ratio increased risk of dying also increased. This supports the idea that omega-3s have stronger effects than omega-6s.

Strengths:

This is a large study (over 85,000 participants) with a good follow up period (average 12.7 years). Using blood levels of fatty acids is superior to using estimated dietary intakes. The authors take account of many variables that could interfere with the findings (confounding variables) - they do this using statistical methods.

Weaknesses:

UKBioBank is not entirely representative of the UK population.

<https://doi.org/10.7554/eLife.90132.2.sa0>

Author Response

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The manuscript offers a commendable exploration into the relationship between plasma omega-6/omega-3 fatty acid ratios and mortality outcomes.

Strengths:

The chosen study design and analytical techniques align well with the research objectives, and the results resonate with existing literature.

Weaknesses:

Lack of information on the selection criteria for participants; 5. The analysis of individual PUFAs is not appropriate; The definition of comorbidities is vague; The rationale of conducting the mediation analysis of blood biomarkers is not given.

Thank you for your insightful feedback and for acknowledging the strengths of our manuscript, particularly regarding the alignment of our study design and analytical methods with our research objectives. Your recognition of how our results resonate with existing literature is greatly appreciated.

Addressing the concerns you've raised:

Selection Criteria for Participants: In the “Methods-Study population” section, we have outlined the exclusion criteria for participant selection. This information provides comprehensive insight into our methodology for selecting the study cohort.

Analysis of Individual PUFAs: We acknowledge your concern regarding the analysis of individual PUFAs due to their inter-correlations in plasma levels. However, the correlations between omega-3% and omega-6% ($r = -0.12$) and between DHA% and LA% ($r = 0.03$) are actually low. Because DHA is one of omega-3 PUFAs, we did not include PUFAs in the same model. Similar considerations apply to LA and omega-6. We believe that exploring the effects of individual fatty acids adds valuable depth to our research. Both DHA and LA have been included in the same model due to their low correlation, with careful adjustments for confounding factors to provide a nuanced understanding of their individual impacts on mortality.

Definition of Comorbidities: The definition of comorbidities, including hypertension, diabetes, and longstanding illness, is elaborated under the Methods section. These conditions were identified through self-reported data collected via the Assessment Centre Environment (ACE) touchscreen questionnaire, allowing us to capture a broad range of chronic conditions as reported by participants.

Rationale for Mediation Analysis: Initially, our approach to mediation analysis included various blood biomarkers available in the UK Biobank database to explore the potential underlying pathways. However, upon considering your feedback regarding the overlap of fatty acids with lipid classes or lipid particles in plasma, we have decided to remove these elements from our mediation analysis.

Reviewer #2 (Public Review):*Summary:*

This study utilized a large sample from the UK Biobank which enhanced statistical robustness, employed a prospective design to establish clear temporal relationships, used objective biomarkers for assessing plasma omega-6/omega-3 ratio, and investigated various mortality causes including CVD and cancer for a holistic health understanding.

Strengths:

The authors used a large sample size, employed a prospective design, and investigated various mortality.

Weaknesses:

Analyzing n-3 and n-6 PUFAs separately might be less instructive. It might not be methodologically sound to treat TG, HDL, LDL, and apolipoproteins as mediators. It's imperative to exercise caution when drawing causal conclusions from the observed correlations. The manuscript might propose potential research trajectories.

We are grateful for your thoughtful analysis of our study's strengths and for your constructive feedback on areas for improvement.

Response to Weaknesses:

Analyzing n-3 and n-6 PUFAs Separately: We recognize the challenge in analyzing n-3 and n-6 PUFAs separately due to their correlations. However, the correlation between n-3% and n-6% in UK Biobank was actually relatively low ($r = -0.12$). We include them in one model to test if both are associated with the outcomes after controlling for the effects of the other. Indeed, both were negatively associated with the mortality outcomes in our analysis. We believe our supplemental analysis of n-3 and n-6 PUFAs provides useful information to the readers, in addition to our findings based on the n-6/n-3 ratio.

Mediation Analysis of TG, HDL, LDL, and Apolipoproteins: We appreciate your insight on the methodological considerations of treating these biomarkers as mediators. After careful review and in line with suggestions from another reviewer, we have removed these elements from our mediation analysis. This revision improves the net scientific rigor of our work, ensuring that our conclusions are drawn from the most robust and methodologically sound of our analyses.

Causal Conclusions from Correlations: We fully agree with the need for caution in interpreting correlations in observational studies. To this end, we have avoided implying causality in our manuscript. Terms suggesting causality, like "protective effects," have been replaced with "inverse associations" to more accurately represent our findings. This adjustment enhances the clarity and accuracy of our conclusions.

Proposing Future Research Trajectories: Recognizing the importance of advancing causal and mechanistic understanding in this field, we have called for future studies to further examine causality and characterize molecular mechanisms of the observed associations in our study.

Reviewer #3 (Public Review):*Summary:*

The authors are trying to find out whether the levels of omega-6 and omega-3 fatty acids in the blood are linked to the likelihood of dying from anything, of dying from cancer and of dying from cardiovascular disease. They use a large dataset called UK Biobank where fatty acid levels were measured in blood at the start of the study and what happened to the participants over the following years (average of 12.7 years) was followed. They find that both omega-6 AND omega-3 fatty acids were linked with less likelihood of dying from anything, from cancer and from cardiovascular disease. The effects of omega-3s were stronger. They then made a ratio of omega-6 to omega-3 fatty acids and found that as that ratio increased risk of dying also increased,. This supports the idea that omega-3s have stronger effects than omega-6s.

Strengths:

This is a large study (over 85,000 participants) with a good follow up period (average 12.7 years). Using blood levels of fatty acids is superior to using estimated dietary intakes. The authors take account of many variables that could interfere with the findings (confounding variables) - they do this using statistical methods.

Weaknesses:

There are several omega-6 and omega-3 fatty acids - it is not clear which ones were actually measured in this study

Thank you for recognizing the strengths of our study, including the large sample size, the duration of follow-up, and our methodological approach to using blood levels of fatty acids and addressing potential confounders. Regarding the weakness you've highlighted, we understand the importance of specifying which omega-6 and omega-3 fatty acids were analyzed in our study. We have revised the Method section to provide detailed information about how the exposures were measured.

Recommendations for the author:

Reviewer #1 (Recommendations for the Authors):

To elevate the manuscript's scholarly rigor, I propose the following refinements:

(1) The manuscript lacks information on the selection criteria for participants and the representativeness of the UK Biobank cohort. It is important to provide details on how participants were selected and whether it is representative of the general population, which is crucial for assessing the generalizability of the findings.

We appreciate the opportunity to clarify the participant selection criteria and the representativeness of the UK Biobank cohort within our manuscript. In the "Methods-Study population" section, we delineated the exclusion criteria: "Participants with cancer (n=37,736) or CVD (n=100,972), those who withdrew from the study (n=879), and those with incomplete data on the plasma omega-6/omega-3 ratio (n=277,372) were excluded from this study, leaving 85,425 participants, 6,461 died during follow-up, including 2,794 from cancer and 1,668 from CVD." To further address representativeness, we performed a sensitivity analysis, examining the baseline characteristics of participants included in our study relative to those omitted due to lack of exposure information. This analysis, presented in Additional file 2: Table S13, indicates comparable baseline characteristics across both participant groups, bolstering confidence in the representativeness of our study sample with the general UK Biobank participants.

Regarding the UK Biobank's representativeness with the general population, we acknowledge that the cohort does not mirror the broader UK demographic in terms of socioeconomic and

health profiles. Participants in the UK Biobank generally exhibit better health and higher socioeconomic status than the average UK resident, potentially influencing the disease prevalence and incidence rates. Nonetheless, the UK Biobank's extensive sample size and comprehensive exposure data enable the generation of valid estimates for exposure-disease associations. These estimates have been corroborated by findings from more demographically representative cohorts, as highlighted in the studies by Batty et al., and Fry et al..

We recognize the importance of this aspect and will incorporate a discussion on the implications of these factors for the generalizability of our findings in the “Discussion-Limitations” section of our manuscript. We are grateful for this insightful comment and believe that this addition will enhance the manuscript's contribution to the field.

Here is what we added in the “Discussion-Limitations” section of our manuscript: “Third, we acknowledged that the cohort did not mirror the broader UK demographic in terms of socioeconomic and health profiles. Participants in the UK Biobank generally exhibited better health and higher socioeconomic status than the average UK resident, potentially influencing the disease prevalence and incidence rates. Nonetheless, the UK Biobank's extensive sample size and comprehensive exposure data enable the generation of valid estimates for exposure-disease associations. These estimates have been corroborated by findings from more demographically representative cohorts^{47,48}.”

References:

Batty, G. D., Gale, C. R., Kivimäki, M., et al. Comparison of risk factor associations in UK Biobank against representative, general population based studies with conventional response rates: prospective cohort study and individual participant meta-analysis. *BMJ*. 2020; 368: m131.

Fry A, Littlejohns TJ, Sudlow C, et al. Comparison of Sociodemographic and Health-Related Characteristics of UK Biobank Participants With Those of the General Population. *Am J Epidemiol*. 2017;186(9):1026–34.

(2) The study sample included different ancestries which may introduce confounding from genetic background. As over 90% of the participants were of European ancestry, I recommend excluding individuals of non-European ancestry in the main analysis.

Thank you for raising the concern regarding the inclusion of different ancestries in our study sample and the potential confounding. In our research, we have adhered to the widely accepted practice of including all participants in the study to ensure a comprehensive analysis. Recognizing the predominance of European ancestry within our cohort, which exceeds 90%, we have proactively incorporated ethnicity as a covariate in our statistical models to mitigate confounding influences.

We also considered the feasibility of conducting a stratified analysis for non-European participants. However, the small sample sizes of non-European subgroups do not provide sufficient statistical power to yield reliable or meaningful separate analyses. Consequently, to maintain the integrity and robustness of our findings, we opted to include all participants in the main analysis, adjusting for ethnicity to account for potential confounders.

(3) I noted that a large proportion of participants were excluded due to the lack of data on plasma PUFAs. Were the characteristics of these participants similar to the current analysis sample?

Thank you for raising this very important point. According to UK Biobank, “The EDTA plasma samples were picked randomly and are therefore representative of the 502,543 participants

in the full cohort.” (As detailed in Julkunen et al.) Moreover, as noted in our reply to comment #1 above, we performed a sensitivity analysis, examining the baseline characteristics of participants included in our study relative to those omitted due to lack of exposure information.

The results of this analysis are detailed in Additional file 2: Table S13. They demonstrate that the baseline characteristics—such as age, gender, ethnicity, socioeconomic status, and lifestyle habits—are indeed similar between the two groups. This similarity supports the representativeness of our analysis sample and suggests that the exclusion of participants without plasma PUFA data does not introduce a bias that would undermine the validity of our study's findings.

References:

Julkunen H, Cichońska A, Tiainen M, et al. Atlas of plasma NMR biomarkers for health and disease in 118,461 individuals from the UK Biobank. *Nat Commun.* 2023 Feb 3;14(1):604. doi: 10.1038/s41467-023-36231-7.

(4) The methods section should include a detailed description of the measurement of plasma omega-6/omega-3 fatty acid ratio. It is important to provide information on the analytical techniques used and any quality control measures implemented to ensure the accuracy and reliability of the measurements. Importantly, were repeated measurements done?

Thank you for raising this important point. The details of the metabolomic profiling have been described in previous UK Biobank publications. In this revision, we added a brief description of the measurement process and provided references to previous publications.

Here is what we added in the “Methods- Ascertainment of exposure” section of our manuscript: “Metabolomic profiling of plasma samples was performed with high-throughput nuclear magnetic resonance (NMR) spectroscopy. At the time of this analysis (15 Mar 2023), UK Biobank released the Phase 1 metabolomic dataset, which covered a random selection of 118,461 plasma samples from the baseline recruitment. These samples were collected between 2007 and 2010 and had been stored in -80°C freezers, while the NMR measurements took place between 2019 and 2020. Detailed descriptions could be found in previous publications about plasma sample preparation, NMR spectroscopy setup, quality control protocols, correction for sample dilution, verification with duplicate samples and internal controls, and comparisons with independent measurements from clinical chemistry assays20-22.”

(5) The analysis of individual PUFAs is not appropriate because plasma levels of these PUFAs, including n-3 PUFAs and n-6 PUFAs, EPA, DHA and AA, are usually correlated. It is hard to differentiate these correlated FAs in Cox model. Whereas the ratio of n-6/n-3 is indeed more comprehensive, and the current analysis demonstrated this ratio as a good marker of mortality. Therefore, the analyses of individual PUFAs can be removed and only focus on the ratio of n-6/n-3.

We resonate with the Reviewer regarding the importance of focusing on the ratio of n-6/n-3. Indeed, the ratio is our focus in this manuscript. We also acknowledge the Reviewer's concern regarding the inclusion of correlated covariates in one statistical model. In that specific analysis, the correlations between omega-3% and omega-6% ($r = -0.12$) and between DHA% and LA% ($r = 0.03$) are relatively low. Additionally, we also checked the model for multicollinearity and found that the variance inflation factors (VIFs) were within acceptable ranges. In the fully adjusted model that included omega-3% and omega-6%, all variables had VIFs below 1.13, with omega-3% at a VIF of 1.06 and omega-6% at a VIF of 1.12. Similarly, in

the model including DHA% and LA%, all variables also exhibited VIFs under 1.13, with DHA% recording a VIF of 1.07 and LA% a VIF of 1.10. Because DHA is one of omega-3 PUFAs, we did not include them in the same model. We did not include LA and omega-6 in the same model, either. Because the ratio has two components and each component is the sum of multiple individual PUFAs, it is natural to ask which component is more important (e.g., omega-6 or omega-3?), which specific fatty acid is driving the effect of omega-3 PUFAs (e.g., ALA? Or the marine omega-3, EPA and DHA?). We received such feedback frequently when we presented our research previously. Therefore, as an effort to address them, we performed analysis of omega-3, omega-6, DHA, and LA. While we understand the complexities involved in differentiating the effects of individual fatty acids in a Cox model, we believe there is intrinsic value in exploring these relationships further. In our analysis, we have attempted to investigate the effects of individual PUFAs on mortality by including both DHA and LA within the same model due to their low correlation, making adjustments to account for confounding factors (As detailed in Additional file 2: Table S9). Our findings indicate significant inverse associations between both DHA and LA with all-cause, cancer, and cardiovascular disease (CVD) mortality. We agree with the Reviewer that the focus of our manuscript should be the ratio, but also hope the Reviewer will agree with us that keeping the results from individual PUFAs will provide additional useful information to the readers.

(6) The definition of comorbidities (including hypertension, diabetes, and longstanding illness) is vague. Please clarify what diseases longstanding illness includes.

We appreciate the request for clarification regarding the definition of comorbidities in our study, including the categorization of longstanding illness. The information regarding longstanding illnesses was obtained via the Assessment Centre Environment (ACE) touchscreen questionnaire. Participants were asked, "Do you have any long-standing illness, disability, or infirmity?" with the response options being "Yes," "No," "Do not know," and "Prefer not to answer." For the purposes of our analysis, participants who selected "Yes" were categorized as having a longstanding illness, while the remaining options were grouped as not having a longstanding illness.

This method of classification aligns with our detailed explanation in the "Methods-Ascertainment of covariates" section of the manuscript, where we state that "Comorbidities, including hypertension, diabetes, and longstanding illness, were self-reported at baseline. Longstanding illness refers to any long-standing illness, disability, or infirmity, without other specific information." It is important to note that this approach is consistent with established precedents in the field. Specifically, the paper by Li et al. in the BMJ utilized a similar definition for comorbidities, reinforcing the validity of our methodology.

References:

Li ZH, Zhong WF, Liu S, et al. Associations of habitual fish oil supplementation with cardiovascular outcomes and all cause mortality: evidence from a large population based cohort study. *BMJ*. 2020 Mar 4;368:m456.

(7) The rationale of conducting the mediation analysis of blood biomarkers is not given. Since fatty acids can be formed as TG or bound with apolipoproteins in plasma, there is a large overlap of FAs with these biomarkers and thus it is not appropriate to analyze TG, HDL, LDL, and apolipoproteins as mediators.

We are grateful for the insightful feedback regarding the mediation analysis of blood biomarkers. Our mediation analysis aimed to explore the possible biomarkers and biological processes that explain the effects of PUFAs on mortality. Upon reflection, we recognize the complexities introduced by the inherent overlap of fatty acids with different lipid particles and lipid classes in plasma. Considering the potential confounding this overlap presents, and

in agreement with your recommendation, we have decided to remove the mediation analyses involving cholesterol, TG, HDL-C, LDL-C, Lp(a), ApoA, and ApoB from our study. We appreciate your guidance on this matter and have updated our manuscript accordingly to reflect these changes.

Reviewer #2 (Recommendations for the Authors):

(1) Analyzing n-3 and n-6 PUFAs separately might be less instructive given the inherent correlations among plasma levels of n-3 PUFAs and n-6 PUFAs. Also, some important specific PUFAs, such as ALA, AA, EPA, etc. were not available in the UK Biobank data though the authors tried to analyze LA and DHA. The n-6/n-3 ratio, as evidenced by the current analysis, offers a more holistic perspective and might be a superior mortality marker. Thus, I recommend shifting the focus solely to this ratio.

Thank you for the thoughtful comment. Reviewer #1 raised a similar point (comment #5 above). We are glad that both reviewers recognized the importance of the omega-6/omega-3 ratio and agreed with us that the ratio should be the focus of the paper. Please also see our more detailed response above. Briefly, our manuscript centered on the ratio, while the supplemental analysis of omega-3%, omega-6%, DHA%, and LA% provided additional useful information. We included omega-3% and omega-6% in the same model because their correlation was relatively low ($r = -0.12$). We also checked the model for multicollinearity and found that the variance inflation factors (VIFs) for n-3 PUFAs and n-6 PUFAs were within acceptable ranges. In the fully adjusted model that included omega-3% and omega-6%, all variables had VIFs below 1.13, with omega-3% at a VIF of 1.06 and omega-6% at a VIF of 1.12. Similarly, in the model including DHA% and LA%, all variables also exhibited VIFs under 1.13, with DHA% recording a VIF of 1.07 and LA% a VIF of 1.10. Therefore, we decided to keep the content for omega-3 and omega-6 PUFAs. We hope that Reviewer will agree with us that this content only provides additional information to the readers.

(2) It might not be methodologically sound to treat TG, HDL, LDL, and apolipoproteins as mediators. Since the model included comorbidities as covariates, hypercholesterolemia and hypertriglyceridemia seemed to have been adjusted in the analysis. Thus, further adjusting these blood biomarkers for mediation analysis which overlapped with comorbidities is redundant.

We appreciate your critical evaluation of our methodological approach. Your point is well-taken, especially in light of the fact that comorbidities such as hypercholesterolemia and hypertriglyceridemia have been accounted for as covariates in our model. This overlap, as you correctly identified, could indeed render the mediation analysis redundant. In concordance with your recommendation, and incorporating the comments of another reviewer, we have now omitted the mediation analysis involving these blood biomarkers from our study. We believe this adjustment strengthens the methodological soundness of our research and are thankful for your contribution to this refinement. We have updated our manuscript to reflect these changes and ensure our analysis remains robust and free from redundancy.

(3) It's imperative to exercise caution when drawing causal conclusions from the observed correlations. The inherent constraints of observational studies, coupled with potential residual confounding or reverse causality, should be acknowledged.

We concur with the caution against implying causality from correlations observed in our study. As such, we have carefully refrained from claiming any causal relationships within our paper. We acknowledge that the term "protective effects" could suggest a causal inference,

and we have revised our language to describe these observations as "inverse associations" to more accurately reflect the nature of our findings.

We have also addressed the inherent limitations of observational research in the Discussion section under 'limitations' of our manuscript. There, we recognize that while we have accounted for many confounders, the possibility of residual confounding cannot be entirely excluded. We also agree that reverse causality is a concern in observational studies. To mitigate this, we performed a sensitivity analysis excluding participants who died within the first year of follow-up. The results from this analysis, which are provided in Additional file 2: Table S12, show consistency with our main findings, suggesting that the observed associations are less likely to be predominantly driven by reverse causation. We are grateful for your insights, which have guided us in strengthening our manuscript and ensuring that our conclusions are presented with the appropriate scientific rigor.

(4) To guide subsequent scholarly endeavors, the manuscript might propose potential research trajectories, such as spearheading randomized controlled trials to delve deeper into the causal nexus between plasma omega-6/omega-3 ratios and mortality outcomes or probing the mechanistic underpinnings of the observed correlations.

We agree that conducting randomized controlled trials could illuminate the potential causal relationships between plasma PUFA biomarkers and mortality outcomes. While the primary focus of our manuscript is to report on associations, we acknowledge the importance of causal analysis in advancing the field. In our secondary analysis, we touched upon mediation effects of blood biomarkers, which could serve as a preliminary step towards establishing causality. Although our current work did not delve deeply into causal mechanisms, the results we have presented may indeed stimulate further exploration. By reporting our mediation analysis results, we aim to provide a foundation that other researchers might build upon. We hope that our work will act as a catalyst for more in-depth studies, such as RCTs or mechanistic investigations, to pursue the questions we have begun to explore.

Following this recommendation, we have revised our Conclusion paragraph and added: "Our findings support the active management of a high circulating level of omega-3 fatty acids and a low omega-6/omega-3 ratio to prevent premature death. Future research is warranted to further test the causality, such as Mendelian randomization and randomized controlled trials. Mechanistic research, including comprehensive mediation analysis, in-depth experimental characterization in animal models or cell lines, and intervention studies, is also needed to unravel the molecular and physiological underpinnings."

Reviewer #3 (Recommendations for the Authors):

(1) Line 32. Delete "a balanced" because a balanced o6:o3 cannot be defined.

Thank you for pointing out the issue with the term "a balanced". Most authors agree with your observation that defining what constitutes a 'balanced' ratio can be ambiguous and potentially misleading. One author, JTB, disagrees that "balance" as a concept is unacceptably ambiguous or misleading. In response, we have removed the words from our manuscript.

(2) In the abstract you should present the findings for omega-6 and omega-3 PUFAs first and then the findings for the ratio.

We appreciate your suggestion to present the findings for omega-6 and omega-3 PUFAs prior to those for the ratio in the abstract. As laid out in the Background section, the ratio was our primary exposure of interest. So, we organized our manuscript by centering on the ratio. We are glad that both Reviewer #1 and #2 expressed a particular interest in the ratio findings and

urged us to keep the ratio as the focus. We believe that this emphasis reflects the novel aspects of our research and aligns with the thematic structure of our manuscript.

| (3) *Line 80. controversial should read uncertain.*

Thank you for the suggestion. We have changed “controversial” to “uncertain”.

| (4) *It is unclear which fatty acids are included in total PUFAs, omega-6 PUFAs and omega-3 PUFAs. It is vital that this is specified.*

Thank you very much for your suggestion. We agree that it is important to clarify the specific fatty acids included in the analysis. In the revised manuscript, we emphasized that we analyzed “total omega-6 PUFAs” and “total omega-3 PUFAs”, while “LA is one type of omega-6 PUFAs” and “DHA is one type of omega-3 PUFAs”. We also revised the Method section of “Ascertainment of exposure” to provide more information about how the exposures were measured. Here is what we added in the “Methods- Ascertainment of exposure” section of our manuscript: “Five PUFAs-related biomarkers were directly measured in absolute concentration units (mmol/L), including total PUFAs, total omega-3 PUFAs, total omega-6 PUFAs, docosahexaenoic acid (DHA), and linoleic acid (LA). Of note, DHA is one type of omega-3 PUFAs, and LA is one type of omega-6 PUFAs. Our primary exposure of interest, the omega-6/omega-3 ratio, was calculated based on their absolute concentrations. We also performed supplemental analysis for four exposures, the percentages of omega-3 PUFAs, omega-6 PUFAs, DHA, and LA in total fatty acids (omega-3%, omega-6%, DHA%, and LA%), which were calculated by dividing their absolute concentrations to that of total fatty acids.”